

Reetesh Mukul

Senior Principal Applied Scientist | Multimodal Learning, Document AI, and Visual Intelligence

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Research Profile

Senior Principal Applied Scientist who owns applied research from problem formulation through model design, training, evaluation, and productization. Focused on large multimodal models, policy verification for agentic AI, visual grounding, document intelligence, and video reasoning, with a research agenda spanning uncertainty-aware learning, efficient routing, and robust visual understanding.

Research and Engineering Focus

Multimodal Learning and Evaluation: vision-language model adaptation, visual grounding, cross-modal alignment, temporal reasoning, and evaluation design. **Agentic AI Guardrails:** LLM-powered policy verification, policy and query intermediate representations, query-to-policy mapping, constraint-aware reasoning, and multi-step action evaluation. **Document Intelligence:** schema-guided extraction, multilingual OCR, complex layouts, reading order, tables, and key-value extraction. **Scientific Learning Methods:** Bayesian uncertainty estimation, causal modeling, reinforcement learning, and experiment-driven model analysis. **Robust Visual Systems:** visual guard models, synthetic-media detection, face anti-spoofing, and mixture-of-experts routing.

Research Highlights

Policy verification for agentic AI. Develops an LLM-powered policy verifier for agentic-AI guardrails, using Policy IR and Query IR, query-to-policy mapping, constraint-aware reasoning, and multi-step action evaluation.

Structured document intelligence. Develops schema-guided LMM workflows for multilingual OCR, complex layouts, reading order, tables, and constrained key-value extraction.

Visual authenticity and video reasoning. Studies multimodal evidence for deepfake detection, face anti-spoofing, and frame-level video analysis, with an emphasis on robust evaluation.

Adaptive visual systems. Combines vision transformers, Bayesian learning, diffusion methods, reinforcement learning, and routing to model memorability, improve creative content, and optimize system behavior.

Experience

Senior Principal Applied Scientist

Dec 2023-Present

Oracle India | Bengaluru

Own the applied-research agenda for large multimodal models, from problem formulation and model design through training, evaluation, and productization. Most recently developed an LLM-powered policy-verification codebase for agentic-AI guardrails, comprising Policy IR, Query IR, query-to-policy mapping, constraint-aware reasoning, and multi-step action evaluation. Also develop schema-guided document intelligence for multilingual OCR, invoices, and tables; and advance biometric deepfake detection and video reasoning for frame-level authenticity analysis.

Principal Machine Learning Engineer

Dec 2022-Nov 2023

Huawei Singapore Research Center | Singapore

Led research-to-prototype work on advertising creative improvement using diffusion-based inpainting and outpainting. Developed visual-content understanding and ad-effectiveness models grounded in attention, saliency, memorability, Bayesian reasoning, and vision transformers.

Principal Machine Learning Engineer

May 2021-Nov 2022

Huawei India Research Center | India

Led research on image memorability and cognitive aspects of visual content, connecting advertising-effectiveness signals to click-through-rate prediction. Applied Bayesian probabilistic programming and trained dense vision-transformer models for image memorability.

Senior Computer Scientist

Sep 2016-Apr 2021

Adobe | India

Led applied ML and performance research for creative applications, including visual similarity, panoptic segmentation, and interactive

systems. Developed reinforcement-learning and multi-armed-bandit methods for adaptive scheduling, resource management, and job-executor control; co-invented techniques for responsive input, UI scheduling, teeth-region detection, and application-performance management.

Senior Software Developer / ML Engineer
Flipkart | India
Developed ad click-prediction models for CTR, CVR, ranking, and sampling, and contributed to the Android native-ad SDK.

2015-2016

Software Developer to Senior Software Developer
Qualcomm | India
Built OCR for Indic Devanagari characters, scene-detection capabilities for camera workflows, voice state-machine components, and embedded drivers.

2008-2015

Software Developer
Texas Instruments | India
Developed software for GPS components in embedded environments.

2006-2008

Selected Patents and Publications

Schema-guided LMM key-value extraction Oracle | US 2026/0127165 A1 | 2026 | Vision-transformer image memorability Oracle | US 2026/0105736 A1 | 2026 | Multimodal video highlight reasoning Oracle | US 2026/0187146 A1 | 2026 | RL-based resource management Adobe | US 11,556,393 B2 | 2023 | Bandit-controlled job-executor policies Adobe | US 12,547,446 B2 | 2026 | WiMAX QoS bandwidth requests IFIP WOCN | 2006

Education

M.Tech, IIIT Bangalore | B.Tech, BIT Sindri